QP CODE - **25901357**

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECJMC201 – UNDERSTANDING PHOTOGRAPHY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks : 50**

PART A — Multiple Choice Questions

1b) To provide a standardized threaded hole for stabilization gear

2c) The Focus ring on the lens barrel

3b) 50mm

4c) Hot Shoe

5b) It lacks an internal reflex mirror and optical viewfinder

6c) Simplify by excluding distractions

7b) bottom third line

8a) darker than the subject and draws attention away

9b) sharp-edged

10b) f/2.0, longer focal length, subject far from background

11a) Three-point lighting

12b) camera leveling/composition control

13a) Three-point lighting

14d) Side lighting

15a) Loop

PART B — Short Answer Questions *(3 Marks)*

16. If you widen your Aperture (e.g., from f/8 to f/2.8) to let in more light, what must you do to your Shutter Speed to maintain the same exposure level? Explain the visual consequence of this change.

- When aperture is widened (e.g., f/8 to f/2.8), significantly more light enters the sensor, so shutter speed must be increased (made faster) to maintain the same exposure level
- The visual consequence is motion freezing — a faster shutter speed reduces or eliminates motion blur in moving subjects
- Additionally, the wider aperture produces a shallower depth of field, creating a blurred background (bokeh) that separates the subject from its surroundings

17. Analyze the relationship between 'Camera Shake' and shutter speed when shooting handheld.

- Camera shake occurs when the photographer's hand movements during handheld shooting cause the camera to move slightly while the shutter is open, resulting in a blurred image

- Slower shutter speeds increase the duration of exposure, giving more time for camera movement to register as blur on the sensor
- The general rule is to use a minimum shutter speed equivalent to 1/focal length (e.g., 1/50s for a 50mm lens) to minimise the risk of camera shake in handheld shooting

18. Analyze the impact of Aperture on 'Lens Speed' and its importance in photojournalism

- Lens speed refers to the maximum aperture a lens can achieve — a fast lens (e.g., f/1.4 or f/1.8) allows significantly lighter to reach the sensor than a slow lens (e.g., f/5.6)
- In photojournalism, a fast lens enables use of faster shutter speeds and lower ISO settings, reducing motion blur and digital noise respectively
- This makes fast lenses essential for photojournalists working in dimly lit environments — courtrooms, indoor events, night scenes — where flash use is often prohibited or impractical

19. Explain why colour helps guide the viewer's attention in a photograph.

- Bright, saturated, or warm colours naturally attract the human eye before neutral or cool tones, directing viewer attention to dominant colour areas within a frame
- Colour contrast — such as a red subject against a green background — creates visual tension that draws the eye immediately to the point of contrast
- Photographers use colour intentionally to establish visual hierarchy, lead the viewer through the composition, and emphasise the primary subject within a busy scene

20. Demonstrate balance- symmetrical or asymmetrical distribution by describing one practical step you would take while framing a shot.

- For symmetrical balance, position the subject at the centre of the frame with equal visual weight on both sides — effective for formal portraits and architectural photography
- For asymmetrical balance, place the main subject off-centre using the rule of thirds, and counterbalance it with a secondary element of lesser visual weight on the opposite side
- A practical step: use the camera's grid overlay to assess weight distribution across the frame before pressing the shutter

21. Design a three-point lighting setup for an interview scenario

- Key light — the primary and brightest light source, placed at approximately 45 degrees to the side and slightly above the subject to illuminate the face and create natural shadow definition
- Fill light — a softer, less intense light placed on the opposite side of the key light to reduce harsh shadows and balance overall exposure on the subject's face
- Back/rim light — positioned behind and above the subject to create a subtle edge highlight that separates the subject from the background, adding depth and a professional, three-dimensional look

22. Produce professional colour correction for digital photographs

- Begin with white balance adjustment to ensure neutral tones — whites appear white and skin tones look natural — using tools in Lightroom, Camera Raw, or Capture One
- Adjust exposure, highlights, shadows, and contrast to achieve a balanced tonal range across the image without clipping highlights or crushing shadows
- Apply targeted HSL (Hue, Saturation, Luminance) corrections to individual colour channels for precise control, then export in the appropriate colour profile (sRGB for web, Adobe RGB for print)

23. Design a flat lighting setup for group photography

- Position a large, diffused light source — a softbox or umbrella — directly in front of and at the same level as the group to minimise shadows across all faces
- Use a second light of equal intensity on the opposite side or a reflector to eliminate any remaining shadow areas, ensuring even illumination across all subjects
- Flat lighting is ideal for group photography as it flatters all skin tones equally, avoids unflattering shadows on individual faces, and ensures uniform exposure across a wide spread of subjects

PART C — Essay Questions *(15 Marks)*

24. You are shooting a fast-moving indoor sport (e.g., basketball) in a poorly lit gymnasium. You find that at ISO 400, your shutter speed is too slow to freeze the players, resulting in blur. Justify the decision to "push" your ISO to 6400. In this scenario, why is a noisy-but-sharp image considered superior to a clean-but-blurry image?

- At ISO 400 with insufficient light, the shutter speed required to freeze fast-moving players will be too slow, producing motion blur that makes the image journalistically and aesthetically unusable
- Pushing ISO to 6400 allows a dramatically faster shutter speed — sufficient to freeze movement — at the cost of increased digital noise (grain) in the image
- A sharp but noisy image retains all critical visual information — player positions, expressions, decisive moments — whereas a clean but blurry image communicates nothing usable to editors or audiences
- In photojournalism, sharpness and moment capture are the primary criteria for publishable images; modern noise-reduction software (Lightroom, Topaz DeNoise) can reduce grain in post-production but cannot recover lost sharpness from motion blur
- The decision to push ISO reflects a professional judgement that technical imperfection is preferable to missing the shot — a core principle of field photojournalism where opportunity cannot be recreated

25. Contrast the internal structural design of a DSLR with a Mirrorless camera. Discuss how the presence or absence of a reflex mirror affects the size, weight, and the way an image is previewed by the photographer.

- A DSLR (Digital Single-Lens Reflex) contains a reflex mirror positioned at 45 degrees behind the lens that reflects incoming light upward through a pentaprism into an optical viewfinder, allowing the photographer to see the actual scene through the lens
- When the shutter fires, the mirror flips up, allowing light to reach the image sensor — this mechanical process creates the characteristic shutter sound and introduces a brief blackout in the viewfinder
- A mirrorless camera eliminates the reflex mirror and pentaprism entirely — light passes directly onto the sensor at all times, which feeds a live electronic viewfinder (EVF) or rear LCD screen
- The absence of the mirror box significantly reduces the camera's physical depth, making mirrorless systems more compact and lighter than equivalent DSLRs, while enabling shorter flange distances that allow new lens designs
- Mirrorless cameras offer real-time exposure preview in the EVF, silent shooting modes, and faster continuous shooting rates, representing the current direction of professional camera development

26. Explain how you would show the true scale of a monument/building in a photograph.

- Include a human figure, vehicle, or familiar object of known size in the foreground or midground of the frame — the viewer's brain uses this reference to calibrate the true scale of the structure
- Shoot from a low angle with a wide-angle lens to emphasise the height and grandeur of the building while keeping the human reference visible in the lower portion of the frame
- Use leading lines — pathways, roads, colonnades — that draw the eye from the foreground reference point toward the monument, reinforcing the sense of distance and scale through perspective

- Avoid telephoto compression, which flattens spatial relationships and makes large structures appear closer to surrounding elements than they actually are
- Compositional placement of the monument filling the frame against a sky with small human figures at its base is a classic technique that communicates overwhelming scale intuitively to the viewer

27. Design a lighting setup for cinematic storytelling

- Key light — use a hard or moderately diffused source positioned at 45–90 degrees to the subject to create dramatic shadow definition that establishes mood and character; lower placement suggests menace, higher placement suggests authority
- Rim/hair light — a strong backlight separates the subject from the background, creating the signature cinematic edge glow that adds three-dimensionality and visual depth to the frame
- Practical lights — incorporate visible light sources within the scene (lamps, candles, screens) to ground the lighting in narrative reality and add motivated, story-consistent illumination
- Background lighting — control the background separately using coloured gels, projected patterns (gobos), or graduated lighting to create visual depth and establish genre, time of day, or emotional tone
- Colour temperature and contrast — use warm and cool contrast between key and fill sides (colour gels on lights) to achieve the stylised, emotionally resonant look characteristic of cinematic lighting that distinguishes it from flat, even illumination